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AI Ethics & Governance

Credit Application Investigation

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Executive Summary

This report presents the results of a technical investigation into a potential fairness concern within an automated credit card approval system used by ABC Banking in Ontario. The investigation was initiated following a complaint filed with the labour board suggesting that applicants residing in certain postal code regions may have experienced disproportionately high rejection rates when compared with applicants from other geographic areas. Because Canadian financial institutions typically do not collect demographic attributes such as race or ethnicity, the analysis focuses on whether geographic indicators – specifically postal codes – may function as proxy variables that indirectly influence credit approval decisions through broader socioeconomic patterns.

The analysis identified noticeable differences in approval outcomes across several postal code regions. While some variation can be explained by legitimate financial risk indicators such as credit score, income level, and existing debt obligations, additional comparisons suggest that geographic indicators may influence approval outcomes in ways that cannot be fully explained by financial characteristics alone. In practice, this raises questions about whether the automated decision model may be learning patterns associated with geographic location rather than strictly evaluating the financial risk of individual applicants.

The dataset and methodology used in this case study are based on a synthetic data approach. While the data itself does not represent real customer applications, the analysis demonstrates how financial institutions can evaluate automated credit decision systems for potential bias using practical fairness monitoring techniques. In this sense, the exercise is less about proving discrimination and more about illustrating how responsible AI governance frameworks can be used to identify patterns that warrant further investigation within automated financial decision systems.

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1. Case Scenario

ABC Banking has received a formal complaint regarding the outcomes produced by its automated credit card approval system. The complaint alleges that applicants residing in certain postal code regions may be experiencing disproportionately high rejection rates compared with applicants from other geographic areas. The concern raised in the complaint is that the system's approval outcomes may vary across regions in ways that are not immediately clear from the financial characteristics of the applicants themselves.

Following the complaint, the institution's internal governance and model review team has been tasked with conducting a technical investigation into the decision outcomes generated by the automated system. The objective of this investigation is to determine whether measurable differences exist across geographic regions and whether those differences appear consistent with the financial indicators used within the model's evaluation process.

When attempting this investigation, several considerations / real world factors must be considered:

1. **Financial Risk Indicators**: Credit card approval decisions are primarily based on financial risk indicators such as credit score, income stability, debt obligations, and payment history. Differences in approval outcomes across regions may therefore reflect legitimate differences in applicant financial profiles.
2. **Geographic Indicators**: Postal codes are used in the dataset to represent applicant location. While geographic variables can provide useful contextual information, they may also correlate with broader socioeconomic conditions that influence model outcomes.
3. **Indirect Variables**: Even when protected demographic attributes are not explicitly used in a model, other variables may indirectly encode similar information. Geographic indicators such as postal codes may function as proxy variables for socioeconomic characteristics.
4. **Data Availability Constraints:** Canadian credit applications typically do not collect demographic attributes such as race or ethnicity. As a result, fairness investigations must rely on indirect indicators when evaluating whether approval outcomes vary across populations.
5. **Model Governance and Monitoring:** Automated decision systems require regular monitoring to ensure that their outcomes remain consistent with institutional governance standards and regulatory expectations.

The purpose of the investigation is therefore to review the approval outcomes produced by the automated system and determine whether geographic patterns within the data warrant further examination within the institution's model governance framework.

2. Historical Context

Concerns surrounding fairness in lending decisions are not new to Canadian financial institutions. Banks have historically been required to monitor lending practices to ensure that credit decisions are based on legitimate financial risk indicators rather than geographic or demographic characteristics. Past regulatory scrutiny has shown that geographic indicators can sometimes reflect broader socioeconomic conditions, which may influence how credit is distributed across communities (FCAC, 2023).

For these reasons, Canadian banks have long maintained internal governance and compliance processes to review lending outcomes and identify patterns that may raise fairness concerns. These reviews often examine approval rates across regions and income groups to ensure that lending practices remain consistent with consumer protection expectations and regulatory oversight (OSFI, 2023). What has changed in recent years is the growing reliance on automated decision systems to evaluate credit applications. Machine learning models are now capable of identifying complex patterns within financial data and processing applications at a much larger scale than traditional rule-based systems. While these systems improve efficiency, they also introduce new governance challenges when model variables correlate with broader socioeconomic patterns.

More recently, the adoption of large language models and other advanced AI systems has accelerated the integration of artificial intelligence into financial decision processes. As institutions continue to adopt these technologies, longstanding concerns surrounding fairness, transparency, and oversight in automated decision systems have re-emerged within modern AI governance discussions.

3. Regulatory Context

Financial institutions that operate in Ontario must ensure that automated decision systems comply with several legal and regulatory frameworks governing consumer protection, privacy, and fair lending practices. These frameworks establish expectations around how customer data is used and how financial decisions are made.

One of the primary federal privacy frameworks is the Personal Information Protection and Electronic Documents Act (PIPEDA). PIPEDA requires organizations to collect and use personal information responsibly and to ensure that automated processes involving customer data are transparent and justifiable. When automated systems are used to assist with financial decision-making, institutions must ensure that the use of personal data remains consistent with the purposes for which it was collected. In addition to federal privacy legislation, financial institutions must also operate in accordance with the Ontario Human Rights Code, which prohibits discrimination in services based on protected characteristics such as race, ethnicity, and place of origin.

Canadian banks are also expected to follow guidance from the Financial Consumer Agency of Canada (FCAC) regarding the fair treatment of financial consumers. These expectations emphasize transparency, responsible use of technology, and oversight of automated processes that affect consumer financial outcomes. Additionally, institutions are required to comply with model governance expectations outlined by the Office of the Superintendent of Financial Institutions (OSFI). OSFI's model risk management guidance requires banks to maintain oversight of models used in financial decision-making, including monitoring model performance and evaluating potential risks associated with automated decision systems.

Together, these frameworks establish the governance environment in which automated credit decision systems must operate. Investigations such as the one described in this case study are therefore conducted not only to evaluate technical model behavior, but also to ensure that automated decision systems remain aligned with regulatory expectations surrounding fairness, accountability, and responsible use of data.

4. Scope of Investigation

Following the complaint, the objective of the investigation is to evaluate whether the automated credit approval system produces materially different outcomes across geographic regions. Because demographic attributes such as race or ethnicity are not collected within Canadian credit applications, the investigation focuses on observable variables that may influence approval decisions. In particular, the analysis examines how approval outcomes vary across postal code regions and whether these differences remain present when applicants share similar financial characteristics.

To conduct this review, the investigation focuses on three primary areas:

1. **Approval Outcomes Across Regions-** Approval and rejection rates are compared across postal code regions to determine whether measurable geographic differences exist within the dataset.
2. **Financial Profile Comparisons-** Applicants with similar financial characteristics (e.g. credit score, income, and debt-to-income ratio) are compared across regions to evaluate whether geographic location influences approval outcomes.
3. **Model Decision Patterns-** The model score and approval threshold are examined to determine how applicants are distributed across approval and rejection decisions.

By isolating the analysis to the three respective areas, the investigation will aim to determine whether geographic differences in approval outcomes appear consistent with financial risk indicators or whether the patterns observed suggest that geographic signals may be influencing model decisions.

5. Methodology

A structured analytical approach was implemented to evaluate whether geographic patterns exist within the automated credit approval system. As access to real customer application data is restricted due to privacy and regulatory considerations, a synthetic dataset was generated to simulate the types of financial attributes typically used in credit evaluation models.

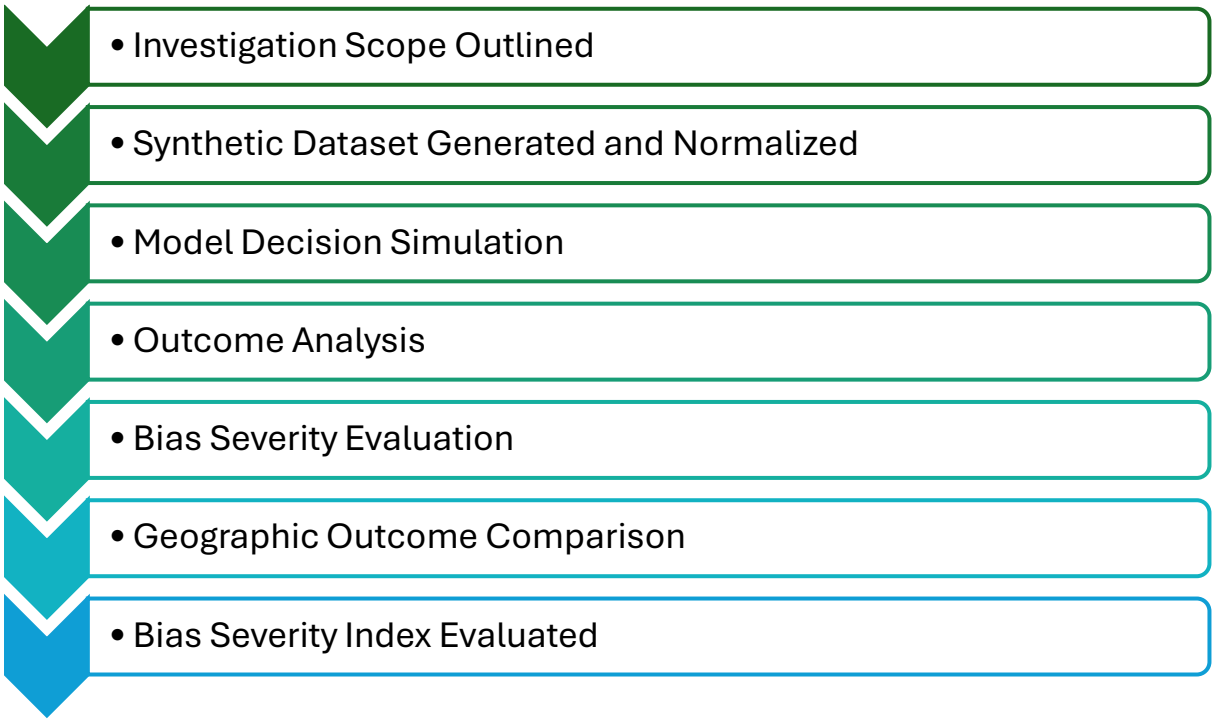


Figure 1. Methodology workflow for the Automated Credit Approval Investigation.

5.1 - Investigation Scope Outlined

This section defines the objective and analytical scope of the investigation. The purpose is to determine whether approval outcomes produced by the automated credit decision system vary across geographic regions and whether those differences can be explained by financial risk indicators or may indicate the presence of unintended bias.

Component	Technical Description
Investigation Objective	Evaluate whether approval outcomes generated by the automated credit model vary across geographic regions after controlling for financial risk indicators.
Target Variable	Credit approval decision (Approved / Rejected).
Geographic Feature	Postal code region used to group applicants into geographic cohorts.
Financial Risk Features	Credit score, annual income, debt-to-income ratio (DTI), credit utilization, missed payment history.
Analytical Constraint	Protected demographic attributes such as race or ethnicity are not collected in Canadian credit applications, requiring indirect fairness evaluation using proxy indicators.
Analytical Outcome	Identify geographic disparities in approval outcomes that persist after accounting for financial features.

5.2 - Synthetic Dataset Generated and Normalized

This section describes the creation of the synthetic dataset used to simulate credit card applications. Because real customer data cannot be used due to privacy and regulatory constraints, a simulated dataset was generated to replicate the financial attributes typically used in automated credit evaluation models.

Dataset Parameter	Technical Description
Dataset Type	Synthetic credit applicant dataset generated using Python.
Record Structure	Each record represents a simulated credit applicant with associated financial attributes.
Sample Size	Simulated population of applicants designed to approximate typical consumer credit application volumes.
Credit Score Distribution	Randomized within a realistic range (typically 300–850).
Income Distribution	Generated within a range representative of consumer income brackets.
Debt-to-Income Ratio	Calculated as total debt obligations divided by annual income.
Credit Utilization	Simulated as the percentage of available revolving credit currently used by the applicant.
Payment History	Number of missed payments recorded within the previous 12 months.
Geographic Attribute	Postal code region assigned to each applicant to enable regional analysis.

5.3 - Model Decision Simulation

This section outlines how automated credit approval decisions were simulated using the synthetic dataset. A simplified scoring model was used to generate approval outcomes based on key financial indicators such as credit score, income level, and debt obligations.

Model Element	Technical Description
Model Type	Simplified credit scoring model used to simulate automated approval decisions.
Model Inputs	Financial variables from the synthetic dataset (credit score, income, DTI, utilization, payment history).
Model Score Calculation	Weighted linear combination of financial risk indicators used to estimate applicant creditworthiness.
Scoring Function	Model Score = Credit Score + Income − DTI − Utilization − Missed Payments.
Approval Threshold	Applications with model scores above a predefined threshold are classified as approved.
Rejection Condition	Applications with scores below the threshold are classified as rejected.
Geographic Adjustment	Selected postal code regions receive a simulated adjustment factor to evaluate the detectability of geographic bias.

5.4 - Outcome Analysis

This section defines the financial and geographic variables included in the analysis. These variables represent the types of attributes commonly evaluated by automated credit decision systems when assessing applicant creditworthiness.

Analytical Procedure	Technical Description
Approval Rate Calculation	Compute approval proportion for each postal code region.
Regional Distribution Analysis	Compare approval outcomes across geographic cohorts.
Income Band Segmentation	Applicants grouped into income bands to evaluate approval differences within financial strategy.
Matched Profile Comparison	Applicants with similar credit score, income, and DTI values compared across regions.
Model Score Distribution	Evaluate how applicants are distributed relative to the approval threshold.
Analytical Objective	Determine whether geographic disparities remain after accounting for financial risk indicators.

5.5 - Bias Severity Evaluation

This section describes how approval and rejection outcomes were generated using the simulated credit scoring model. The resulting decisions form the basis for evaluating whether patterns in approval outcomes exist across different geographic regions.

Metric Component	Technical Description
Metric Name	Bias Severity Index (BSI).
Purpose	Quantify deviations in approval outcomes across geographic regions relative to the baseline approval rate.
Baseline Calculation	Overall approval rate produced by the model across all applicants.
Regional Measurement	Compute approval rate for each postal code region.
BSI Formula	BSI = | Regional Approval Rate − Baseline Approval Rate |.
Interpretation	Higher BSI values indicate regions where approval outcomes deviate significantly from expected patterns.
Governance Use	Identify geographic cohorts that may require further model investigation within the institution's model governance framework.

5.6 Geographic Outcome Comparison

This section compares approval outcomes across postal code regions in order to identify whether geographic disparities exist within the automated credit approval process. Approval rates are calculated for each region and visualized using bar charts to highlight differences in approval patterns across geographic groups.

Following the generation of approval outcomes from the simulated credit decision model, approval decisions are grouped by postal code region in order to evaluate whether geographic patterns exist within the model's outcomes. Because demographic attributes such as race or ethnicity are not collected in Canadian credit applications, postal codes are used as a geographic proxy to examine whether certain regions experience disproportionately high or low approval rates.

The approval rate is calculated using the following expression:

$$\boldsymbol{Approval\ Rate} = \frac{\text{Approved Applications}}{\text{Total Applications in Region}}$$

This metric will allow the investigation to compare the proportion of approved applicants across the respective regions. They will be then visualized using a bar chart, where each bar represents the approval rate for a given region.

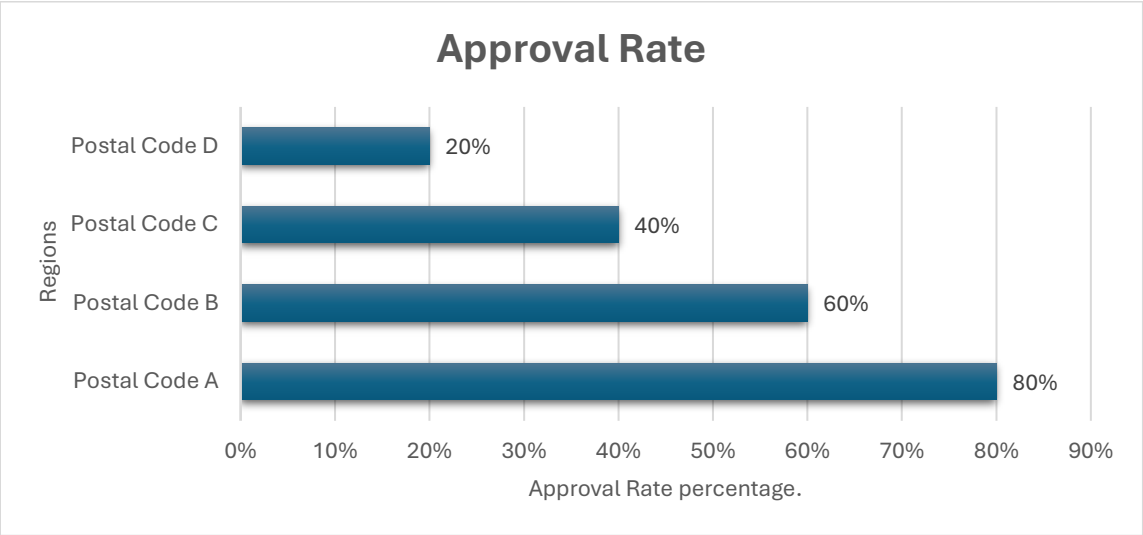


Figure 2- Credit approval rate by Postal Code Region

5.7 Bias Severity Index Evaluation

A quantitative metric is required to measure the magnitude of those differences. To accomplish this, a **Bias Severity Index (BSI)** is calculated for each region. The BSI measures how far a region's approval rate deviates from the overall approval rate produced by the model.

The baseline approval rate across the dataset is first calculated:

$$\boldsymbol{Baseline\ Approval\ Rate} = \frac{\text{Total Approved Applications}}{\text{Total Applications}}$$

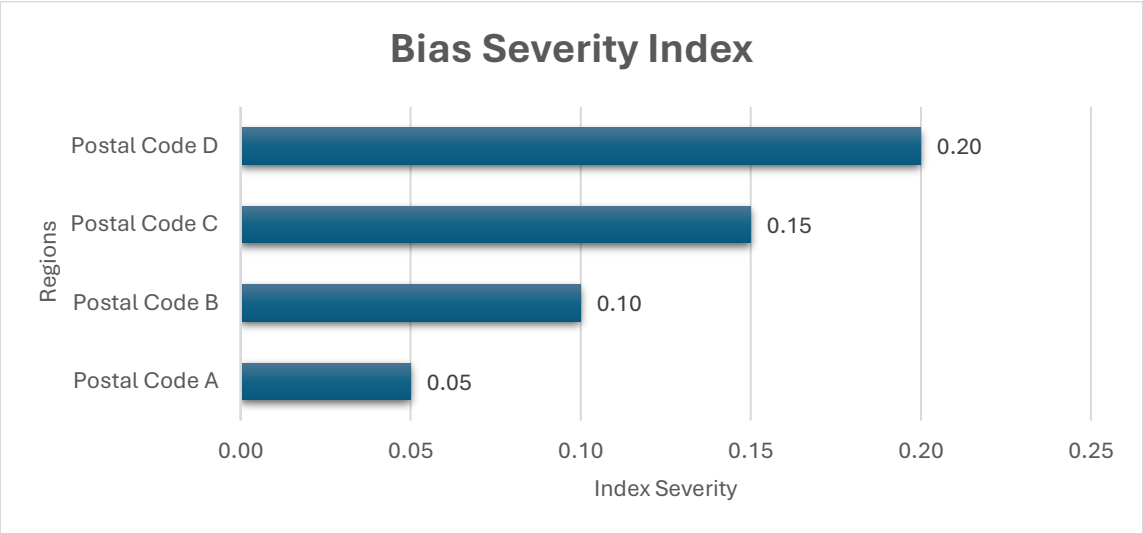
The Bias Severity Index is then computed using the following expression:

$$\boldsymbol{Bias\ Severity\ Index} = |\,\mathrm{AR_r} - \mathrm{AR_{baseline}}|$$

Where:

1. Regional Approval rates represent the approval rate for a specified region.
2. Baseline Approval rates represent the overall approach rate for the entire dataset.

The results are then visualized using a bar chart. Where the height of each bar represents the magnitude of the deviation from the baseline approval rate.



Regions with higher BSI values indicate greater deviation from the model's expected approval pattern and may represent areas where geographic indicators influence model outcomes.

6. Results

The following section presents the outputs produced by the simulated credit decision model described in the methodology. The analysis evaluates approval outcomes across financial characteristics and geographic regions to determine whether measurable disparities exist within the automated credit approval process. The results are presented across three primary analyses: **model decision distribution, geographic approval rate comparison, and Bias Severity Index evaluation.**

6.1 – Model Decision Distribution

The first visualization examines how the simulated model separates applicants based on financial characteristics.

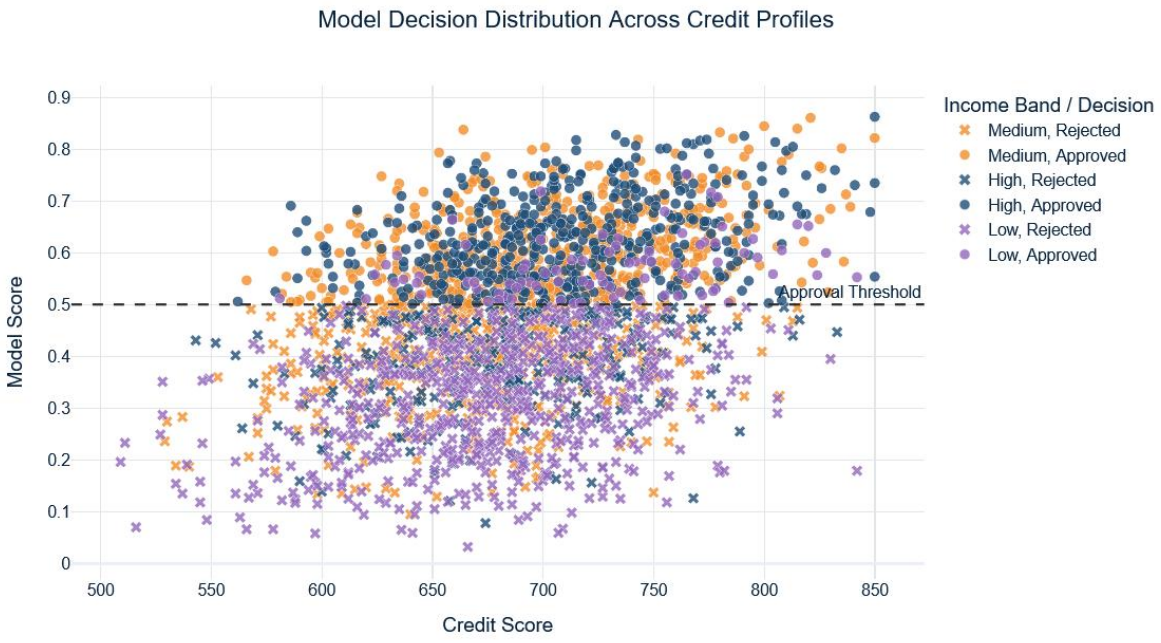


Figure 3 - Model Decision Distribution Across Credit Profiles

The model approves applicants with higher credit scores and model scores, which aligns with standard credit risk evaluation. However, applicants from lower-income regions appear more frequently below the approval threshold, suggesting that geographic characteristics may influence approval outcomes.

6.2 – Approval Rate by Postal Code

The following visualization compares approval rates across postal code regions in order to identify geographic differences in approval outcomes.

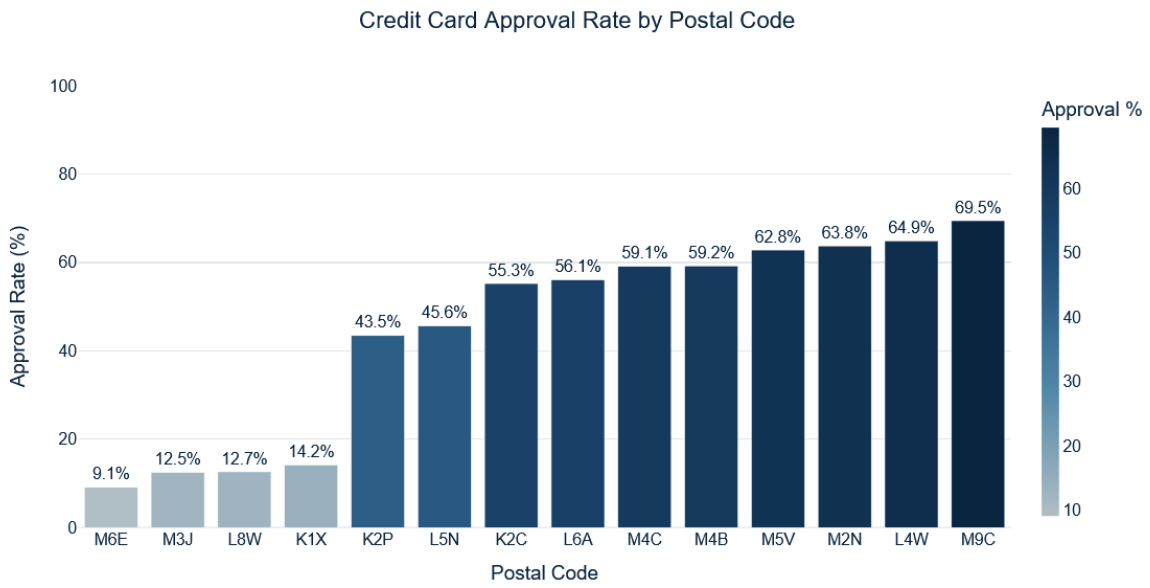


Figure 4 - Credit Card Approval Rate by Postal Code

Approval rates vary significantly across postal codes, with several regions showing approval rates below 15 percent while others exceed 60 percent. This indicates that applicants in certain geographic regions experience different approval outcomes.

6.3 - Approval Rate by Income Band

This visualization aggregates approval outcomes by regional income band to determine whether broader socioeconomic patterns exist within the dataset.

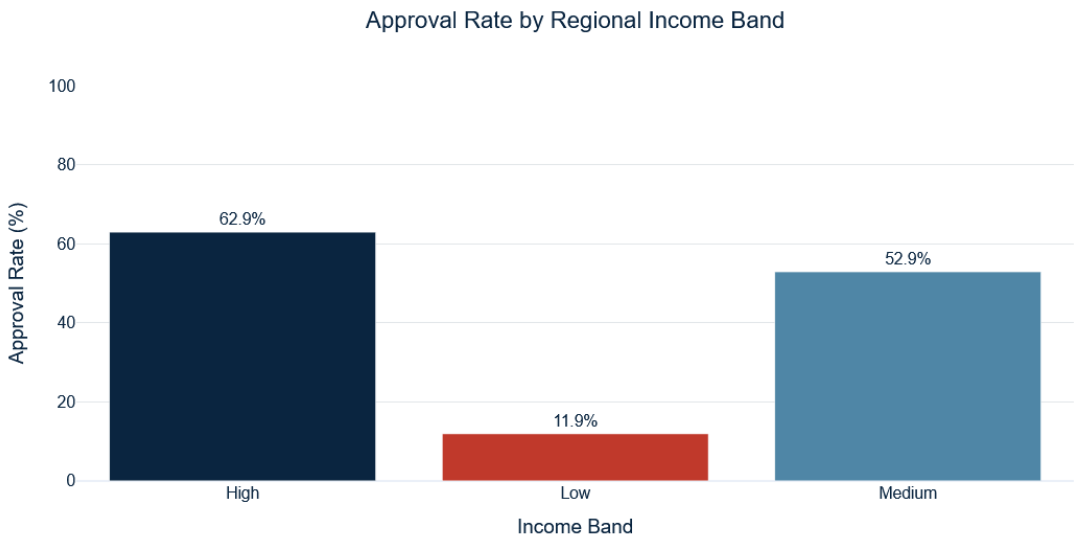


Figure 5- Approval Rate by Regional Income Band

Approval rates increase consistently across income bands, rising from approximately 11.9 percent in low-income regions to over 60 percent in high-income regions.

6.4 - Disparate Impact Ratio

The following visualization evaluates fairness using the Disparate Impact Ratio across regional income groups.

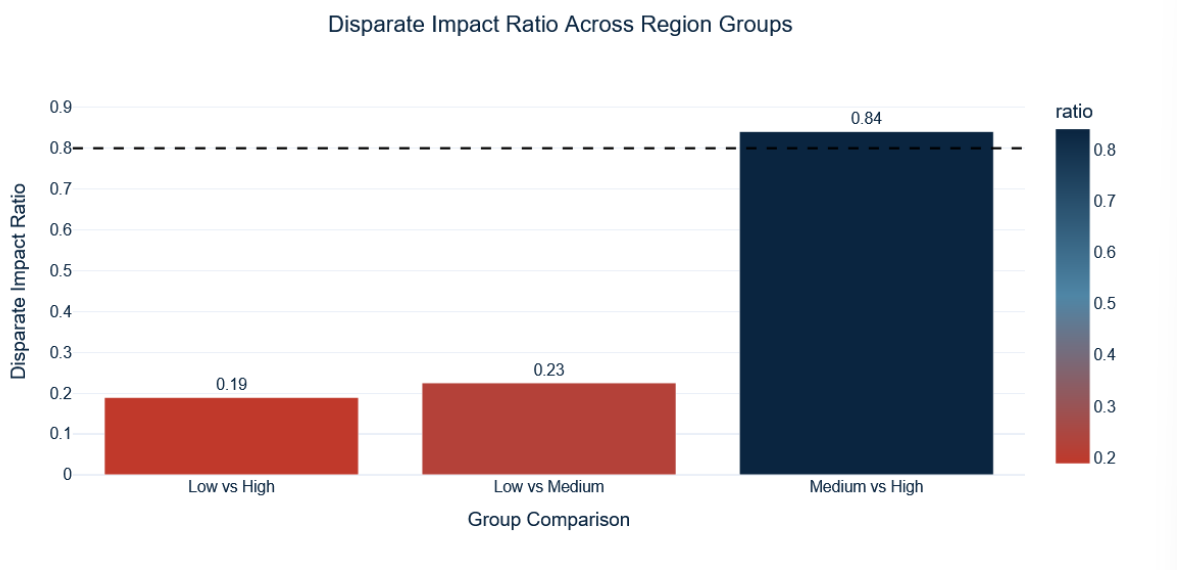


Figure 6 - Disparate Impact Ratio Across Region Groups

The Disparate Impact Ratios for low-income regions fall well below the commonly referenced 0.80 threshold, indicating a substantial disparity in approval outcomes relative to higher-income regions.

6.5 - Bias Severity Index Evaluation

This visualization measures how far each region's approval rate deviates from the baseline approval rate of the model.

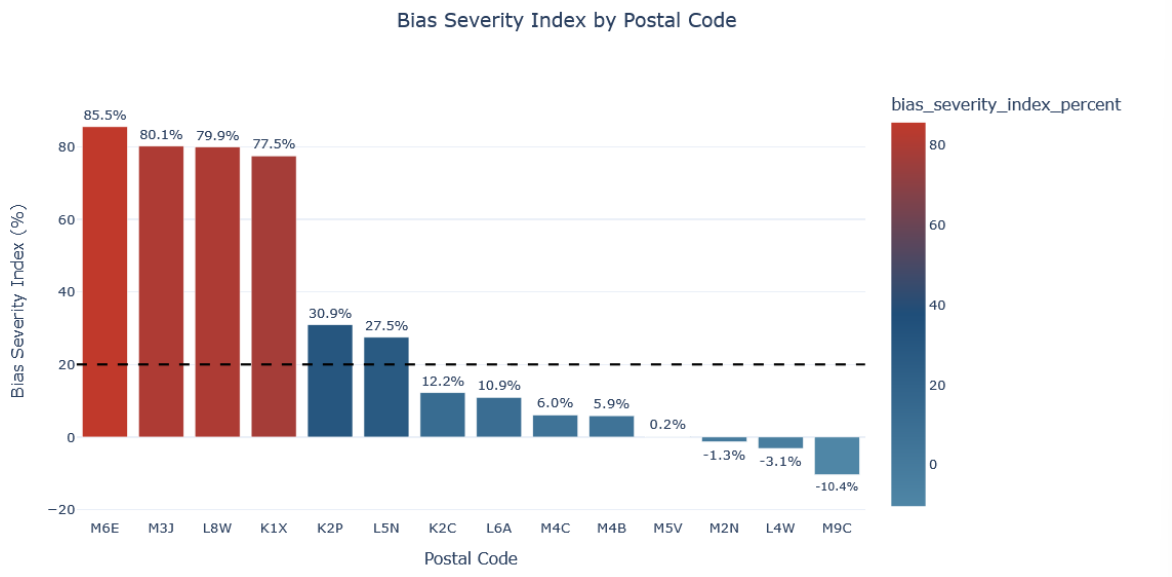
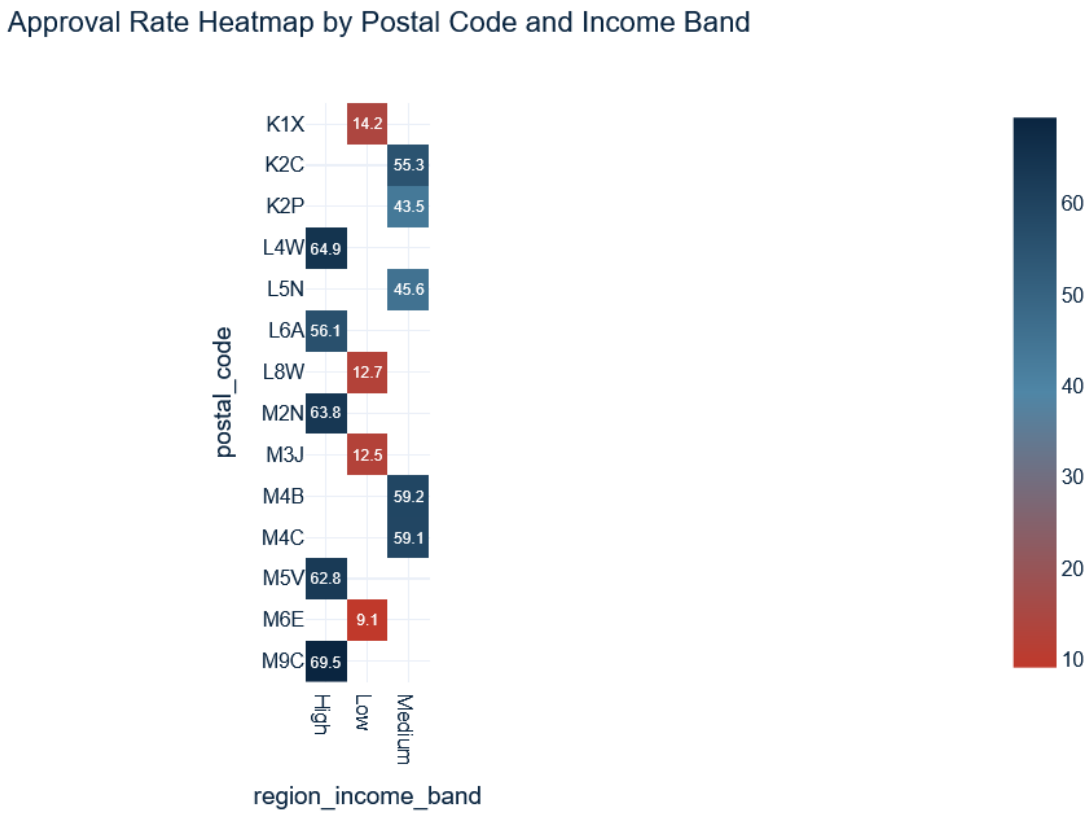


Figure 7 - Bias Severity Index by Postal Code

Several postal codes show extremely high Bias Severity Index values, indicating that approval outcomes in these regions differ significantly from the model's baseline approval rate.

6.6 - Geographic Outcome Comparison

The following visualization compares approval rates across both postal codes and regional income bands to identify broader geographic patterns.



The heatmap highlights a clear pattern where postal codes associated with lower-income regions consistently show lower approval rates than medium- and high-income regions.

7. Governance Conclusions

The results of this investigation suggest that geographic indicators may play a larger role in automated credit decisions than initially expected. Several postal codes associated with lower-income regions demonstrate approval rates that are dramatically lower than those observed in higher-income areas. While some variation in approval outcomes is expected due to differences in financial risk indicators, the magnitude of the gap suggests that geographic variables may influence model behaviour in ways that extend beyond traditional credit risk factors.

The matched comparisons further reinforce this concern. Applicants with broadly similar financial characteristics experienced quite different approval outcomes depending on the region associated with their postal code. When applicants with comparable credit scores, income levels, and debt obligations are treated differently based on geographic location, it raises an important governance question: whether the model is evaluating financial risk, or indirectly learning socioeconomic patterns embedded in the data. This dynamic is not unique to modern machine learning systems. Historically, lending discrimination often occurred through geographic exclusion practices such as redlining in the Jim Crow south. While contemporary credit models no longer rely on explicit demographic variables, geographic features can still function as indirect signals that reproduce similar patterns if they are not carefully monitored.

For financial institutions, the implication is straightforward. Automated decision systems must be evaluated not only for predictive performance, but also for how their outcomes are distributed across different communities. Monitoring geographic approval patterns, identifying potential proxy variables, and regularly auditing model outputs are necessary steps to ensure that automated decision systems remain aligned with fairness expectations and regulatory standards.

8. Conclusion

This investigation demonstrates that the simulated credit approval system produces materially different outcomes across geographic regions. Applicants from various postal codes associated with lower-income communities experienced lower approval rates, even when financial characteristics were broadly comparable. These results illustrate how geographic indicators can function as indirect signals that shape automated credit decisions. As financial institutions continue to rely on automated models, it becomes essential to monitor how those systems distribute outcomes across communities to ensure that credit decisions reflect genuine financial risk rather than patterns embedded within the data itself.

9. References

Financial Consumer Agency of Canada. (2023). Fair treatment of financial consumers framework. Government of Canada. https://www.canada.ca/en/financial-consumer-agency/services/industry/fair-treatment-financial-consumers.html

Office of the Privacy Commissioner of Canada. (2019). Personal Information Protection and Electronic Documents Act (PIPEDA). Government of Canada. https://www.priv.gc.ca/en/privacy-topics/privacy-laws-in-canada/the-personal-information-protection-and-electronic-documents-act-pipeda/

Office of the Superintendent of Financial Institutions. (2023). Guideline E-23: Model risk management. Government of Canada. https://www.osfi-bsif.gc.ca/en/guidance/guidance-library/guideline-e-23-model-risk-management

Ontario Human Rights Commission. (2022). Ontario Human Rights Code. Government of Ontario. https://www.ohrc.on.ca/en/ontario-human-rights-code

Treasury Board of Canada Secretariat. (2020). Directive on automated decision-making. Government of Canada. https://www.tbs-sct.canada.ca/pol/doc-eng.aspx?id=32592